RES-5 endpoint 2PI bound (v1.1, status-reconciled): the endpoint tail is

TECT verification-first repository · claim B1-RH-ENUM

first issued 2026-06-09 · this version issued 2026-06-10 · v1.1

1. Purpose and scope

res5-tail-budget-closure localised RES-5 to the $I = 2 \times 10^{-3}$ endpoint. This note brackets the endpoint between the conservative and verified floor inputs of the SC-SCOPE certified joint and identifies the rigorous residual. **v1.1 correction**: v1.0 mis-stated SC-SCOPE as a live B1 named hypothesis. Canonically (B1 status.json + scscope-programme-consolidation v1.0) SC-SCOPE is LIFTED@THIN-CERTIFIED via the QUARTIC route (Parseval-pinned $R_{\max} = 0.385 < 0.634$); B1 is T6 CONDITIONAL on {H-LAYER} ALONE. The floor route's standalone lift was retracted; the floor ρ_{lat} survives only as the conservative/verified INPUT to the certified joint. Scope: the $I = 2 \times 10^{-3}$ endpoint; estimate / verified grade as marked. No tier action.

2. The SC-SCOPE certified joint and its floor input

The endpoint third-cumulant slack $1 - 1/\mathrm{joint}$ is set by the SC-SCOPE CERTIFIED joint (scscope-quartic-normalisation-certificate), which takes the floor ρ_{lat} as a conservative/verified input:

$$\begin{aligned} \text{conservative } (K_{\mathrm{floor}} \le T') :&\quad \rho_{\mathrm{lat}} = 6.55, \quad \text{joint} = 1.040, \quad \text{slack} = 0.0385, \\ \text{verified } (K_{\mathrm{floor}} \le 0.52\, T' \text{ sample}) :&\quad \rho_{\mathrm{lat}} = 12.6, \quad \text{joint} = 1.082, \quad \text{slack} = 0.0758. \end{aligned}$$

The inequality $K_{\mathrm{floor}} \le T'$ (Parseval + weighted Lemma A) is PROVED; the factor below 1 is realized, not yet a worst-case theorem (Section 4).

3. The tail: operator-norm vs projection

The endpoint tail $C_{\mathrm{higher}}/\Delta F_{\mathrm{margin}} \approx C_G a_0$, $C_G = 1/(1+g) = 0.492$, $a_0 = 0.096$, is an OPERATOR-NORM upper bound. The actual pattern-dependent difference carries a projection $\chi_{\mathrm{proj}} \le 1$,

$$\frac{C_{\mathrm{higher}}}{\Delta F_{\mathrm{margin}}} = \chi_{\mathrm{proj}}\, C_G a_0,$$

with $\chi_{\mathrm{proj}} < 1$ strictly: the source $\lambda'(P^2 - \langle P^2 \rangle)$ is common-mode-subtracted (R-U10-3) and supported on the {110} shell, not the softest screened mode.

4. The floor-κ lever (route A) and its DR-2 character

The lever tightening the joint from conservative ($\kappa = 1$) toward verified is $\kappa = K_{\text{floor}}/T'$, $K_{\text{floor}} = \sum_{t\neq 0} w_t^2/I^2$, $w_t = \sum_{u+v=t} A_u A_v$. Two rigorous results:

(a) **PROVED refinement.** By Cauchy-Schwarz $w_t^2 \leq r(t)S_t$, $S_t = \sum_{u+v=t} A_u^2 A_v^2$, with $\sum_{t\neq 0} S_t = I^2 - \|A\|_4^4$,

$$\boxed{K_{\text{floor}} \leq T'\Big(1 - \tfrac{\|A\|_4^4}{I^2}\Big) \leq T'\big(1 - \tfrac{1}{n}\big).}$$

(b) **EXACT scan.** Over all crystallographic single shells ($n \leq 42$, uniform amplitudes) the exact additive energy gives worst $\kappa = K_{\text{floor}}/T' = 0.75$ (at $R = 4$, $n = 6$) – correcting the prior incomplete-sample 0.52 (which omitted the small shells $R \leq 8$).

The refinement (a) is weak for dense patterns ($1 - 1/n \to 1$); the tight worst-case $\kappa < 1$ over the DENSE admissible class is the additive-energy / circle-incidence problem = the DR-2 frontier. So the floor lever is DR-2-adjacent.

5. Verdict: bracketed; the endpoint unifies with DR-2, not SC-SCOPE

$$\boxed{\text{slack}_{\text{proved}}(0.0385) \ < \ \text{tail}\,(0.047) \ < \ \text{slack}_{\text{verified}}(0.0758)}$$

The endpoint closes at STRONG EVIDENCE by either lever: (i) the verified floor end (38% margin, tail/slack = 0.62), or (ii) the projection $\chi_{\text{proj}} < 0.0385/0.047 = 0.82$. The conservative-grade endpoint ($\kappa = 1$, slack 0.0385) remains marginal.

Corrected unification. SC-SCOPE is lifted@thin-certified (the quartic residual is PINNED); it is NOT a B1 named hypothesis. The RES-5 endpoint's rigorous closure is gated by the worst-case floor $\kappa < 1$, which is the DR-2 additive-energy frontier – the SAME frontier as B1's unrestricted-class question. RES-5's last analytic piece therefore unifies with DR-2, not with SC-SCOPE.

Tier. RES-5/GAP-2: ENDPOINT-LOCALISED $\to$ STRONG EVIDENCE (off-endpoint $\geq 27\times$; endpoint at the verified floor / $\chi_{\text{proj}} < 0.82$). No tier flip: B1-RH-ENUM stays T6 CONDITIONAL on {H-LAYER}. Estimate-grade: $\text{slack}_{\text{verified}}$ and χ_{proj} are realized/physical, not proved worst-case. The rigorous T6 endpoint needs a DR-2-class bound $\kappa < \kappa_{\text{needed}} \approx 0.8$ over the dense admissible class, OR $\chi_{\text{proj}} \leq 0.82$.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *"v1.0 called this 'unifies with SC-SCOPE'; which is right?"* UPHELD against v1.0: SC-SCOPE is lifted (quartic-pinned), so it is not a residual to unify with. The floor κ lever is DR-2-adjacent; v1.1 corrects the unification to DR-2. The numerical bracket is unchanged.

(β) *"The exact* $\kappa = 0.75$ *exceeds the sampled* 0.52 – *does the verified-floor closure survive?"* VALID: with $\kappa = 0.75$ the verified slack is thinner than the 0.52 sample implied, but $0.75 < 1$ still lifts ρ_{lat} above the conservative 6.55; the verified-end closure is retained at STRONG EVIDENCE, and the honest worst-case is the DR-2 bound. The $0.52 \to 0.75$ correction is recorded.

(γ) *"Then the endpoint is still not a theorem."* ACCEPTED: no theorem is claimed; STRONG EVIDENCE only. The advance is (1) the marginalit-y is a conservative-floor artefact, (2) the rigorous residual is identified as DR-2 (one frontier, shared with the unrestricted-class question), and (3) the PROVED $(1-\nu)$ refinement plus the EXACT $\kappa = 0.75$ correction.

Quantitative sanity check (mandatory): *bracket* $0.0385 < 0.047 < 0.0758$; *verified margin* $1 - 0.047/0.0758 = 38\%$; *projection threshold* $0.0385/0.047 = 0.82 < 1$; *exact scan* worst $\kappa = 0.75 \leq 1 - 1/6 = 0.83$ (consistent with the proved $1 - 1/n$ at $n = 6$).

7. Result footer

```
Result ID:                B1-RH-ENUM / RES-5 endpoint 2PI bound (v1.1, reconciled)
Precise statement:        At the I=2e-3 endpoint the operator-norm tail C_G a0=0.047
                          is BRACKETED: slack_proved(0.0385) < tail < slack_verified
                          (0.0758), the slack set by the SC-SCOPE CERTIFIED joint
                          (1.040 conservative floor / 1.082 verified floor). Closes
                          at STRONG EVIDENCE via the verified floor (38% margin) or
                          chi_proj<0.82. The rigorous lever is the floor kappa=
                          K_floor/T': PROVED K<=T'(1-||A||_4^4/I^2); EXACT single-
                          shell worst kappa=0.75 (corrects the 0.52 sample); worst-
                          case kappa<1 over the dense class = DR-2 additive-energy
                          frontier. RES-5 endpoint unifies with DR-2, NOT SC-SCOPE.
Scope:                    I=2e-3 endpoint; STRONG EVIDENCE / verified grade as
                          marked. Conservative-grade endpoint stays marginal.
Dependencies:             res5-tail-budget-closure (localisation); scscope-quartic-
                          normalisation-certificate (CERTIFIED joint 1.040/1.082;
                          SC-SCOPE lift); scscope-programme-consolidation v1.0
                          (canonical: B1 {H-LAYER}); res5-a0-skeleton-sensitivity
                          (C_G, a0); R-U10-3 (common-mode subtraction). Supersedes
                          v1.0 (stale SC-SCOPE-named-hypothesis framing).
Evidence grade:           ANALYTIC (bracket + (1-nu) proof) + EXECUTED
                          (res5_endpoint_2pi_bound.py 5/5) + INHERITED (certified
                          joint).
Reproduction command:     python codes/vacuum/res5_endpoint_2pi_bound.py
Expected output:          tail 0.047 in [0.0385, 0.0758]; tail/slack_verified 0.62;
                          chi_proj threshold 0.82; exact worst kappa 0.75; 5/5 PASS.
Falsification gate:       A DR-2 bound proving kappa>=0.8 worst-case AND chi_proj>0.82
                          (endpoint robustly open); or the certified joint failing.
Tier before / after:      No tier flip. B1 T6 on {H-LAYER}. RES-5/GAP-2: ENDPOINT-
                          LOCALISED -> STRONG EVIDENCE; rigorous T6 endpoint pending
                          the DR-2 kappa bound OR chi_proj<=0.82.
No-overclaim statement:   No theorem. SC-SCOPE is lifted@thin-certified (quartic-
                          pinned), NOT a named hypothesis (v1.0 mis-stated this). The
                          endpoint closes at STRONG EVIDENCE; its rigorous residual is
                          DR-2 (additive-energy), shared with B1's unrestricted-class
                          question. B1 unchanged (T6 on {H-LAYER}).
Next required action:     DR-2: a worst-case additive-energy bound kappa<~0.8 over the
                          dense admissible class (closes the RES-5 endpoint AND the
                          unrestricted-class B1), OR chi_proj<=0.82 ({110}-shell
                          projection of the screened response).
```